

Markscheme

Mathematics: analysis and approaches

Practice question bank

54. (a)

Consider the equation $5^x = 2^{x+3}$.

Solve the equation for x , giving your answer correct to 3 significant figures.

$$\text{take } \ln \text{ of both sides: } x \ln 5 = (x + 3) \ln 2 \quad (M1)$$

$$x(\ln 5 - \ln 2) = 3 \ln 2 \quad \mathbf{A1}$$

$$x = \frac{3 \ln 2}{\ln 5 - \ln 2} \quad (M1)$$

$$x = 2.27 \quad \mathbf{A1}$$

[4 marks]

(b)

Hence find the value of 10^x , correct to 3 significant figures.

$$10^{2.2695 \dots} \quad (M1)$$

$$\approx 186 \quad \mathbf{A1}$$

[2 marks]